

Hospital Readmission Analysis

Linking CMS hospital performance with county socioeconomic context

7,944

OBSERVATIONS

2,444

HOSPITALS

1,197

COUNTIES

23%

HOSPITAL ICC

Khanh (Ada) Nguyen

Applied Statistics & Data Science | R | Healthcare Analytics

The signal behind the headline

55% of hospital-condition records exceeded the CMS benchmark of 1.0.

SCALE

7,944

hospital-condition records

COVERAGE

2,444

hospitals in 1,197 counties

MODEL

23%

residual variance between hospitals

ADJUSTED COMMUNITY ASSOCIATIONS

Poverty

+0.0082

ERR per 1 SD

95% CI: +0.0049 to +0.0116

Unemployment

+0.0069

ERR per 1 SD

95% CI: +0.0045 to +0.0094

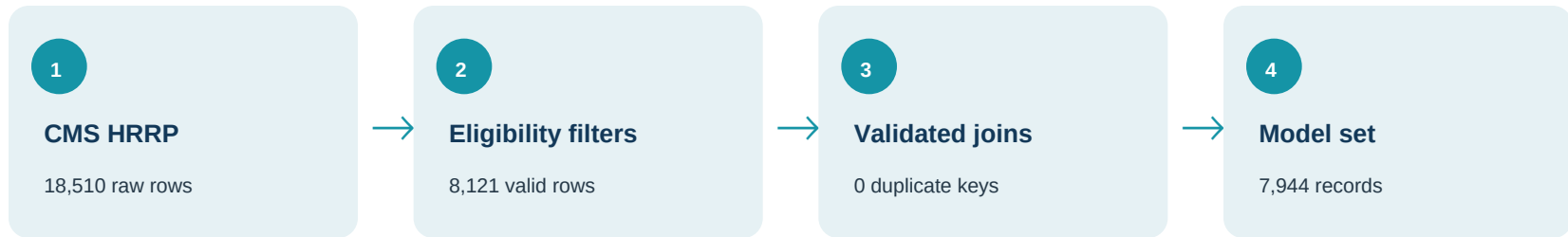
Log population

+0.0060

ERR per 1 SD

95% CI: +0.0034 to +0.0086

From fragmented public data to a validated model



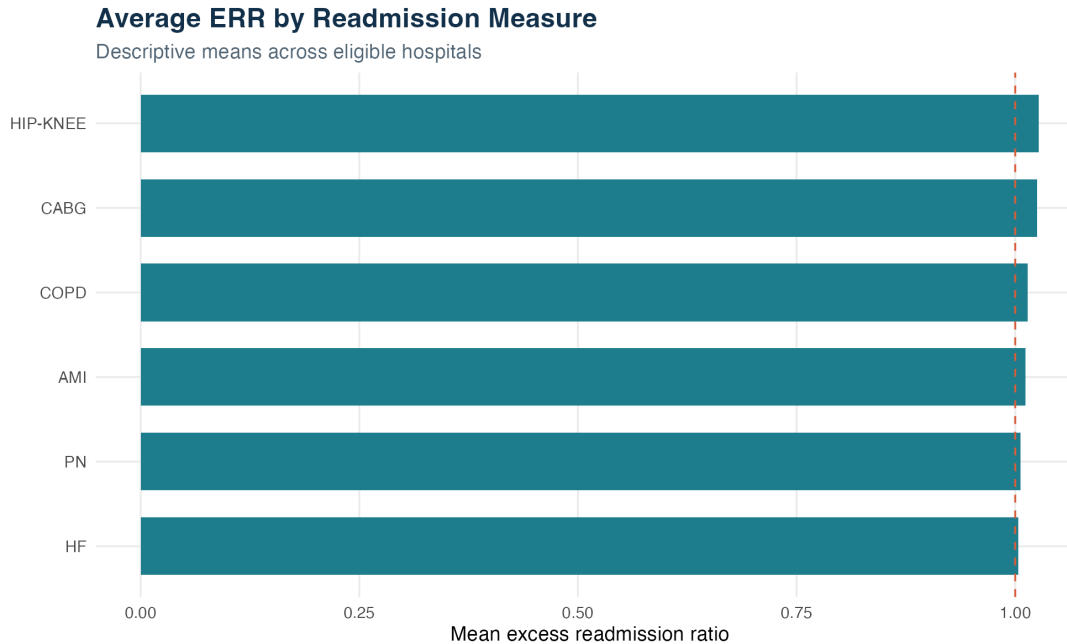
THE CRITICAL FIX

A many-to-many ZIP-to-county join created 2,694 duplicated hospital-measure keys.

The revised pipeline uses state + county name as the primary match and accepts ZIP only when it maps uniquely. County FIPS coverage reached 98.9%, while all 8,121 valid source rows were preserved.

FINDING 1

Condition mix changes the readmission picture



WHAT STANDS OUT

HIP-KNEE and CABG had the highest mean ERR.

CABG also had the largest share above 1.0: 59.9%. COPD followed at 58.6%.

Model implication

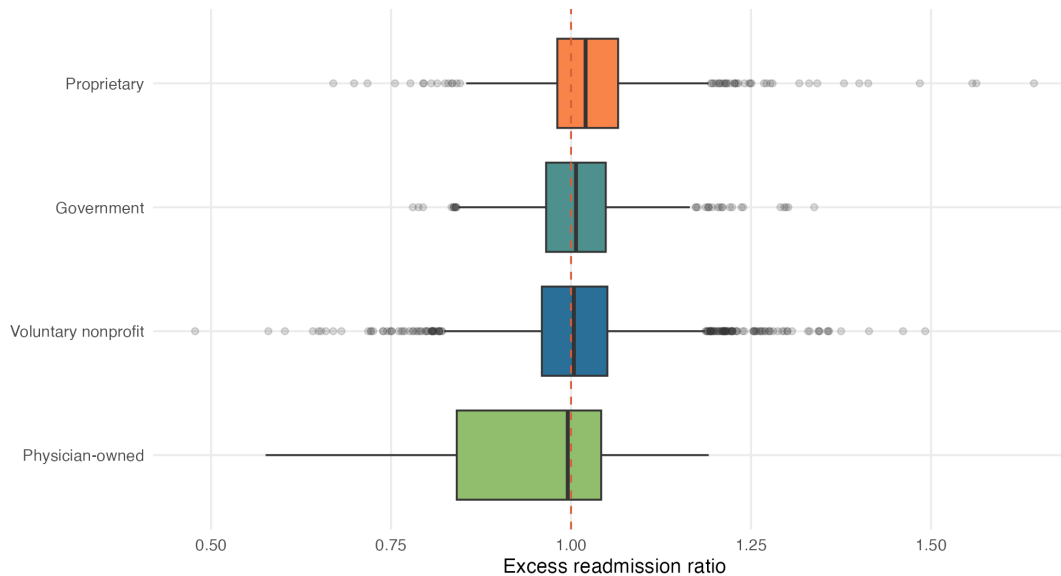
Condition is included as a fixed effect so community comparisons are not driven by the measure mix.

FINDING 2

Ownership differences persist after adjustment

ERR Varies Across Hospital Ownership Groups

Unadjusted distributions; adjusted estimates appear in model results



DESCRIPTIVE RESULT

Proprietary hospitals had the highest mean ERR:

1.0267

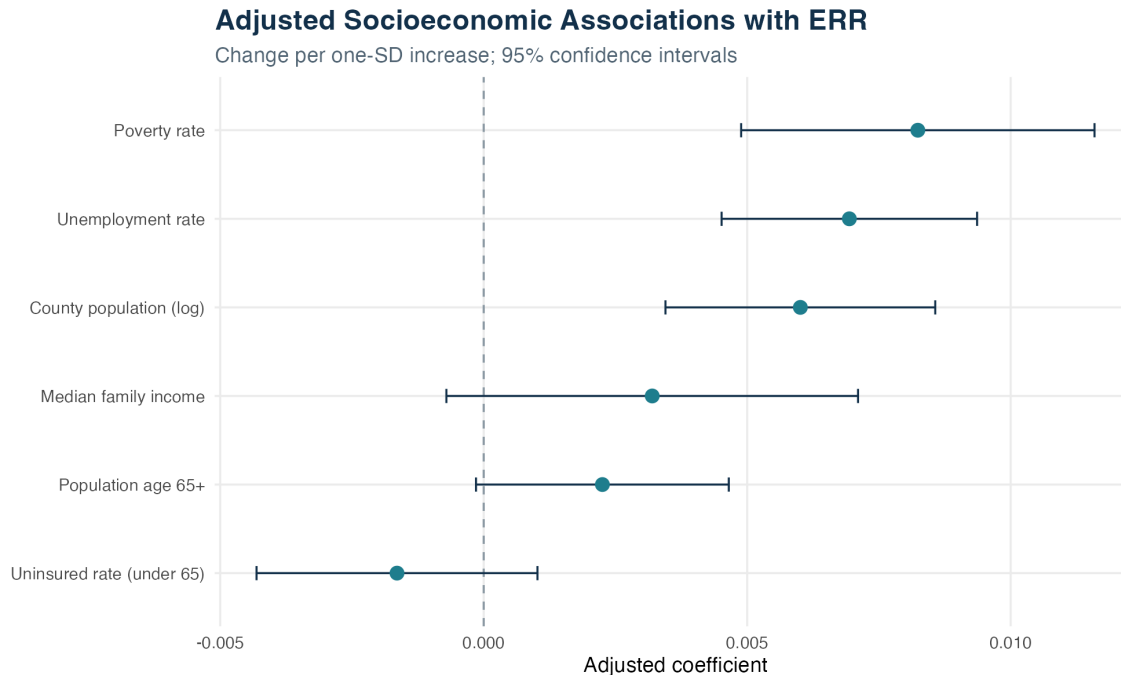
64.7% of proprietary observations were above 1.0.

ADJUSTED RESULT

Physician-owned hospitals showed lower adjusted ERR, but only 26 hospitals were represented.

FINDING 3

Three community measures remained positively associated



ADJUSTED FINDINGS

Poverty

+0.0082 ERR / SD

Unemployment

+0.0069 ERR / SD

Log population

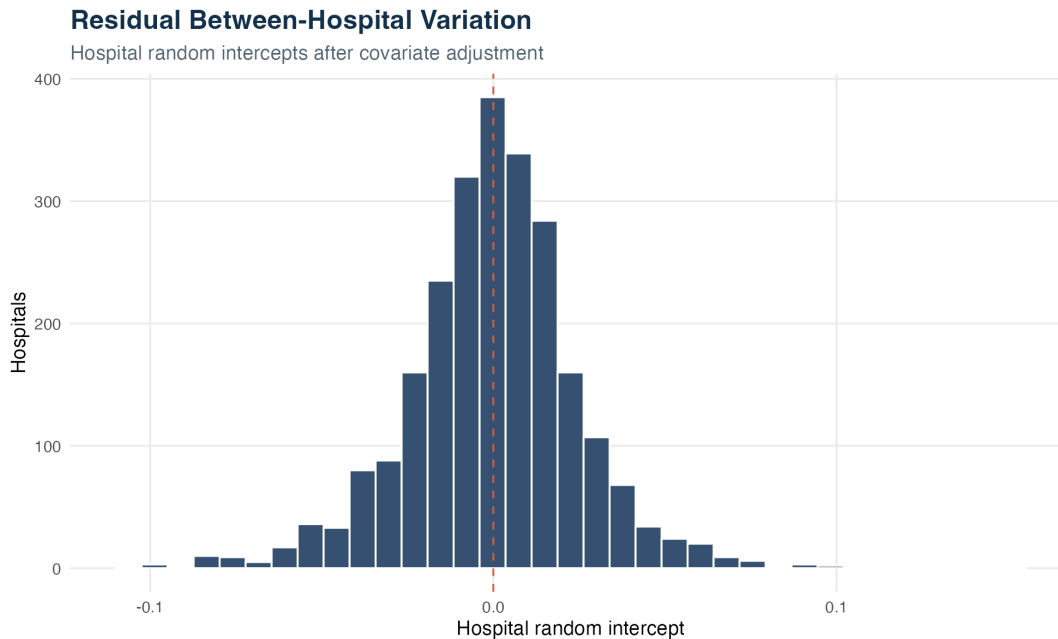
+0.0060 ERR / SD

Income, age 65+, and uninsured rate had confidence intervals that crossed zero.

Points are fixed-effect estimates; horizontal lines are 95% confidence intervals. County covariates are standardized.

FINDING 4

Hospital identity still explains meaningful variation



HOSPITAL-LEVEL ICC

23.0%

Residual similarity within the same hospital

WHY IT MATTERS

Repeated condition records from one hospital are not independent.

The random intercept captures this clustering and prevents overly confident standard errors.

Hospital random-intercept estimates illustrate remaining between-hospital heterogeneity after adjustment.

Khanh (Ada) Nguyen

A reproducible analysis built for decision support

MODEL

- Linear mixed-effects model (nlme)
- Hospital random intercept
- ML comparison; REML final fit

QUALITY CONTROLS

- Key uniqueness checks
- Row-preservation audit
- FIPS coverage and completeness

DELIVERABLES

- Complete R pipeline
- Excel executive report
- Five publication-ready visuals

LIMITATIONS

Observational and ecological analysis; county measures do not represent individual patients. Data years differ across sources, some records lack complete county covariates, and residual confounding remains.

PORTFOLIO TAKEAWAY

Readmission outcomes reflect both hospital-level variation and measurable community context.